Commuting at Pomona: A Dashboard for Sustainable Transportation Options

1st Outline

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1 Introduction

This section will introduce the problem statement and goals of the Sustainability Office as described by the 2030 SAVE goals established. Some details will include metrics such as faculty and staff commuting making up 23% of Pomona's transportation emissions (3% faculty, 20% staff), and the college having 2030 goals of a 30% reduction in SOV commuting miles and a 100% reduction in net commuting emissions.

The goals of the project are a dashboard that helps employees see sustainable commute options from their place of residence, especially since this helps the sustainability office see where reductions are possible. In the introduction, I will also introduce the methodology of the dashboard and the various sources I scrape the public transit data from since there are various entities in Southern California.

2 Background

Additional context will be necessary including Pomona's 2026 Sustainability Annual Report, and the existing incentives (free Foothill Transit with the Class Pass, up to $100/month Metrolink reimbursement, and $2/day for alternative modes) which exist to encourage pulblic transit use, as well as the commute survey the data comes from. I will introduce the GTFS feeds and OpenTripPlanner that will be used in the project and define technical terms like SOV miles, etc so that the average reader can understand the more specific terms used in the project.

3 Data

This section describes the FY26 faculty and staff commute survey. It will have what variables are, how the mode codes work, and how I cleaned and combined the two sheets. I will list the specifics of the sample such as the fact that it is a sample and not the full workforce (112 of 355 faculty and 355 of 820 staff responded), and miles to work is self-reported. I will also describe the external data I included: transit schedules and stops, and ZIP code boundaries.

4 Ethical Considerations

This section covers the ethical issues and how I handled them. The biggest one is privacy since there are many ZIP codes with only one or two respondents. This is why I group areas together and set a minimum group size before showing anything on a map to ensure staff privacy for an area that may be quite distant from campus and identify staff locations. The workbook also contains names and emails in older sheets, which I'll exclude, and ensure the data stays out of public view. I also discuss who the survey might leave out, and that staff who live far away may do so because of housing costs, so the dashboard should present options without blaming people who have few alternatives.

5 Exploratory Analysis

This section shows the main commuting patterns. Faculty mostly live close to campus and already walk or bike often, while staff live farther away and mostly drive alone. I will show shares by distance and compare faculty and staff. I will also show that most SOV miles come from long staff commutes, which shapes where the 30% goal can realistically come from. There will be more analysis in this section as more is learned.

6 Transit and Alternatives Analysis

This section looks at what alternatives exist from each area with nearby transit stops and areas where several employees live close together and could carpool. These results are what the dashboard shows.

7 The Dashboard

This section explains how the dashboard works and how it was built. Employees can enter their area and see their options, travel times, and incentives, and the sustainability office can test scenarios showing how much different changes would move the college toward its 30% goal for further planning.

8 Conclusion

This section summarizes the findings and what they mean for the sustainability office, along with the limitations that exist which includes a small survey sample and no commute start times which can impact travel time due to traffic. I will end with suggestions for future work, like adding questions to the survey about whether employees know about the incentives and how likely staff would be to reconsider their commuting patterns.